3.3.5 Alcohols

Alcohols have many scientific, medicinal and industrial uses. Ethanol is one such alcohol and it is produced using different methods, which are considered in this section. Ethanol can be used as a biofuel.

3.3.5.1 Alcohol production

Content	Opportunities for skills development
Alcohols are produced industrially by hydration of alkenes in the presence of an acid catalyst. Ethanol is produced industrially by fermentation of glucose. The conditions for this process. Ethanol produced industrially by fermentation is separated by fractional distillation and can then be used as a biofuel. **Students should be able to:** • explain the meaning of the term biofuel • justify the conditions used in the production of ethanol by fermentation of glucose • write equations to support the statement that ethanol produced by fermentation is a carbon neutral fuel and give reasons why this statement is not valid • outline the mechanism for the formation of an alcohol by the reaction of an alkene with steam in the presence of an acid catalyst • discuss the environmental (including ethical) issues linked to decision making about biofuel use.	**AT a, d and k** **PS 1.2** Students could produce ethanol by fermentation, followed by purification by fractional distillation.

3.3.5.2 Oxidation of alcohols

Content	Opportunities for skills development
Alcohols are classified as primary, secondary and tertiary. Primary alcohols can be oxidised to aldehydes which can be further oxidised to carboxylic acids. Secondary alcohols can be oxidised to ketones. Tertiary alcohols are not easily oxidised. Acidified potassium dichromate(VI) is a suitable oxidising agent. **Students should be able to:** • write equations for these oxidation reactions (equations showing [O] as oxidant are acceptable) • explain how the method used to oxidise a primary alcohol determines whether an aldehyde or carboxylic acid is obtained • use chemical tests to distinguish between aldehydes and ketones including Fehling's solution and Tollens' reagent.	**AT b, d and k** Students could carry out the preparation of an aldehyde by the oxidation of a primary alcohol. Students could carry out the preparation of a carboxylic acid by the oxidation of a primary alcohol.

3.3.5.3 Elimination

Content	Opportunities for skills development
Alkenes can be formed from alcohols by acid-catalysed elimination reactions. Alkenes produced by this method can be used to produce addition polymers without using monomers derived from crude oil. **Students should be able to:** • outline the mechanism for the elimination of water from alcohols.	**AT b, d, g and k** **PS 4.1** Students could carry out the preparation of cyclohexene from cyclohexanol, including purification using a separating funnel and by distillation.
Required practical 5 Distillation of a product from a reaction.	